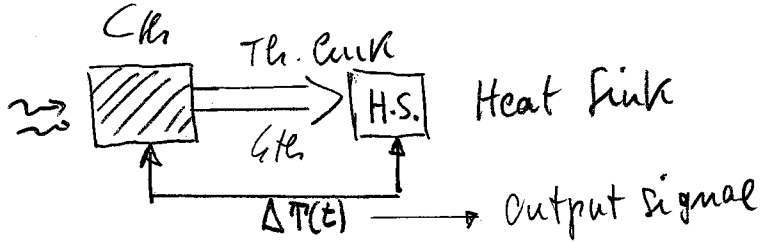
Thermal detectors



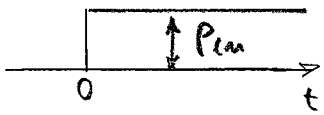
Energy balance in a time interval dt

$$P_{in}\,dt - G_{th}\,\Delta T(t)\,dt = C_{th}\,d[\Delta T(t)]$$

$$P_{in} - G_{th}\,\Delta T(t) = C_{th}\,\frac{d\,\Delta T(t)}{dt}$$

$$\rightarrow \frac{P_{in}}{C_{th}} - \frac{G_{th}}{C_{th}}\,\Delta T(t) = \frac{d\,\Delta T(t)}{dt}$$

P_{in} constant



$t \to \infty$: $\dfrac{P_{in}}{C_{th}} - \dfrac{G_{th}}{C_{th}}\,\Delta T_\infty = 0 \qquad \Delta T_\infty = \dfrac{P_{in}}{G_{th}}$

Hom.: $\dfrac{d\,\Delta T(t)}{dt} + \dfrac{G_{th}}{C_{th}}\,\Delta T(t) = 0 \qquad \Delta T(t) = \Delta T_0 \exp(-t/\tau)$

$$\boxed{\tau = \frac{C_{th}}{G_{th}}}$$

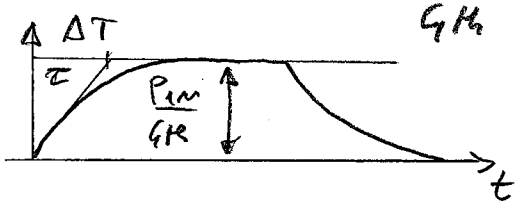
Haeviside fuction

$$\Delta T(t) = \Delta T_0 \exp(-t/\tau) + \frac{P_{in}}{G_{th}}$$

$\Delta T(0) = 0 \rightarrow 0 = \Delta T_0 + \dfrac{P_{in}}{G_{th}} \qquad \Delta T_0 = -\dfrac{P_{in}}{G_{th}}$

IN. COND

Solution $\Delta T(t) = \dfrac{P_{in}}{G_{th}}\left[1 - \exp(-t/\tau)\right]$

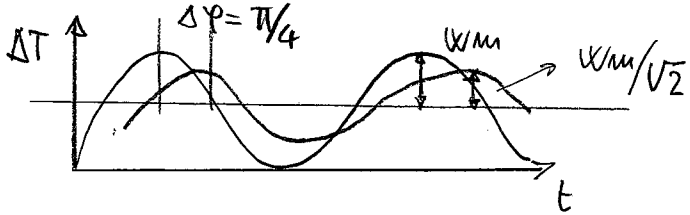


Sinusoidal regime

$P_{in} = W_0 + W_m \sin(2\pi f t) \qquad W_m / W_0 = m \rightarrow$ modulation factor

$m = 1$

$P_{in} = W_m + W_m \sin(2\pi f t)$



$$\Delta T(t) = \frac{W_m}{G_{th}} + \frac{W_m \sin(2\pi f t - \Delta\varphi)}{\sqrt{1 + 4\pi^2 f^2 \tau^2}}$$

$$\Delta T(t) = \frac{W_m}{G_{th}} + \frac{W_m \sin(2\pi f t - \pi/4)}{\sqrt{2}}$$

$$\Delta T^{RMS} = \frac{W_m}{\sqrt{2}} \cdot \frac{\sin(2\pi f t)}{\sqrt{1 + 4\pi^2 f^2 \tau^2}}$$

Cut off frequency

$$f = f_c = \frac{1}{2\pi\tau}$$

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Bolometers [Langley 1878]

Detectors based on change in electrical resistance because of temp. variation

sens. range X-Ray → far IR [No cooling is required]

$R = R_0 (1 + \alpha \Delta T)$ linear model $\alpha = \frac{1}{\rho}\frac{\partial \rho}{\partial T}$ ρ = resistivity

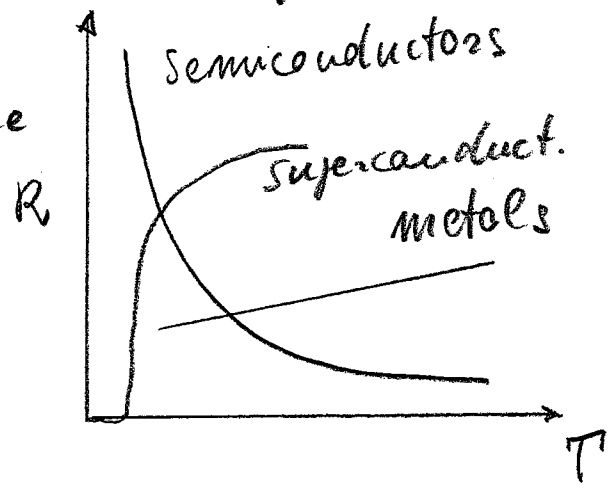
Metals $\alpha \simeq +5 \cdot 10^{-3}\, K^{-1}$

Semiconductors $\alpha \simeq -6 \cdot 10^{-2}\, K^{-1}$

$np = N_c N_v \exp\left(-\frac{E_g}{KT}\right)$ $\begin{cases} N_c = 2\sqrt{(2\pi KT\, m_n^*/h^2)^3} \\ N_v = 2\sqrt{(2\pi KT\, m_p^*/h^2)^3} \end{cases}$

Best sensitivity with high doped semiconductor at low Temperature

NEP → NETD ≃ 50 mK Noise equivalent Temperature difference



Used for thermal IR and THz

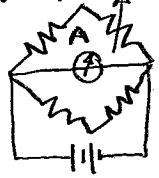
Microbolometers → Thermal imaging

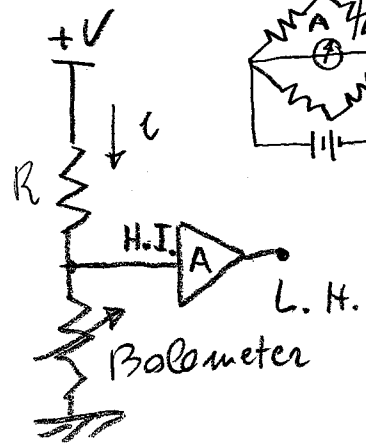
No cooling is required

Superconducting Nanowire Single Photon Detectors SNSPDs

Thermal interaction → Quantum effect

Wheatstone Bridge



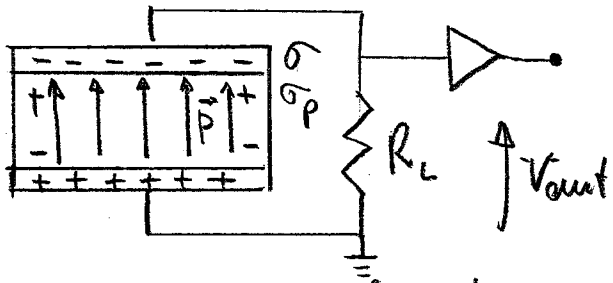


Golay cell [Golay 1947]

Pneumatic detector: the radiation heats a gas (Xenon) in a chamber. The deformation of a membrane is sensed by an optical leverage mechanism [four quadrants P.D.]

Pyroelectric detectors

Based on Ferroeletric materials exhibiting spontaneous polarization



$\sigma_p = \vec{P} \cdot \vec{u}_n \qquad \sigma = -\sigma_p \qquad |\sigma| = |\vec{P}|$

$Q = A\sigma$

The polarization is a function of temperature and time $|\vec{P}(T(t))|$

$$Q(T(t)) = A \cdot |\vec{P}(T(t))| \rightarrow i(t) = \frac{dQ(T(t))}{dt} = A \frac{d|P(T(t))|}{dt}$$

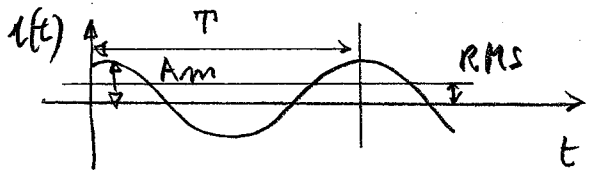
$$i(t) = A \frac{d|P(T)|}{dT} \cdot \frac{dT(t)}{dt} \qquad \frac{d|P(T)|}{dT} = p \rightarrow \text{pyroeletric coefficient}$$

$$i(t) = A\, p \frac{dT(t)}{dt}$$ If P_{in} is constant $\frac{d\Delta T}{dt} = 0 \rightarrow$ No CW Response

Response to a sinusoidal stimulus $\boxed{m = 1}$

$$P_{IN} = W_m + W_m \sin(2\pi f t) \rightarrow \Delta T(t) = \frac{W_m}{G_{th}} + \frac{W_m \sin(2\pi f t)}{G_{th}\sqrt{1 + 4\pi^2 f^2 \tau_{th}^2}} \qquad \tau_{th} = \frac{C_{th}}{G_{th}}$$

$$i(t) = A\,p \cdot \frac{W_m \cdot 2\pi f \cos(2\pi f t)}{G_{th}\sqrt{1 + 4\pi^2 f^2 \tau_{th}^2}}$$



$$RMS = \sqrt{\frac{1}{T}\int_0^T A_m^2 \sin^2(2\pi f t)\,dt}$$

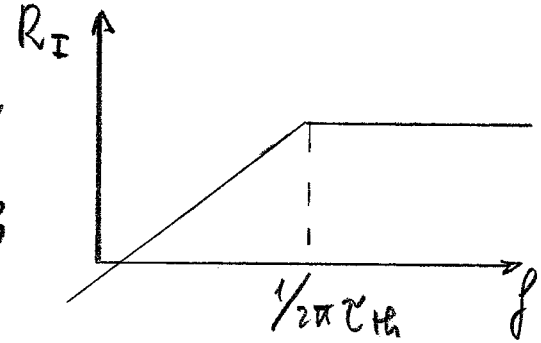
$$T = 2\pi \quad RMS = A_m \sqrt{\frac{1}{2\pi}\int_0^{2\pi} \sin^2(t)\,dt} = \frac{A_m}{\sqrt{2}}$$

$$i^{RMS} = \frac{A \cdot P \cdot W_m \, 2\pi f}{G_{th}\sqrt{2}\,\sqrt{1+4\pi^2 f^2 \tau_{th}^2}} \qquad p^{RMS} = \frac{W_m}{\sqrt{2}} \qquad \tau_{th} = \frac{C_{th}}{G_{th}}$$

The current responsivity is then:

$$R_I^{RMS} = \frac{i^{RMS}}{p^{RMS}} = \frac{A \cdot P \cdot 2\pi f}{G_{th}\sqrt{1+4\pi^2 f^2 \tau_{th}^2}}$$

The resp. has a zero in ϕ and a pole at $1/2\pi\tau_{th}$



For frequencies $f > 1/2\pi\tau_{th}$:

$$R_I^{RMS} = \frac{AP}{G_{th}\tau_{th}} = \frac{A \cdot P}{C_{th}}$$

Equivalent circuit for voltage output

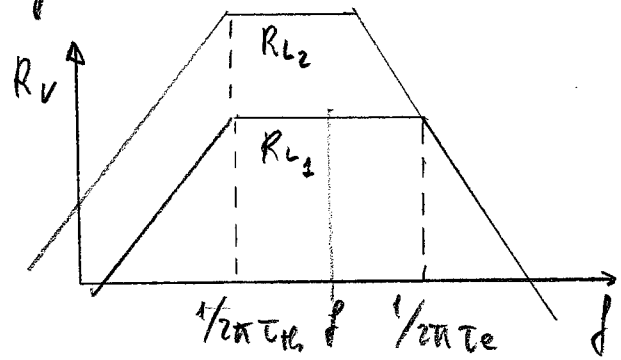
For the Voltage responsivity we must consider an electric pole with $\tau_e = R_L C_e \Rightarrow |H(j\omega)| = \frac{R_L}{\sqrt{1+\omega^2\tau_e^2}}$

$$R_V^{RMS} = R_I^{RMS} \cdot \frac{R_L}{\sqrt{1+4\pi^2 f^2 \tau_e^2}} = \frac{A \cdot P \cdot 2\pi f}{C_{th} C_e} \frac{\tau_{th}}{\sqrt{1+4\pi^2 f^2 \tau_{th}^2}} \cdot \frac{\tau_e}{\sqrt{1+4\pi^2 f^2 \tau_e^2}}$$

$\boxed{\tau_e < \tau_{th}}$ $\quad \frac{1}{2\pi\tau_{th}} < f < \frac{1}{2\pi\tau_e}$

$$\left[R_V^{RMS} = \frac{A \cdot P \cdot R_L}{C_{th}} \right]$$

$$\boxed{\begin{array}{l} \tau_e > \tau_{th} \\ R_V^{RMS} = \dfrac{A \cdot P}{G_{th} C_e} \end{array}}$$



$C_{th} = c_{th}^{vol} \cdot Vol = c_{th}^{vol} \cdot A \cdot d$ $\quad C_e \propto \frac{\varepsilon_r}{d} \Rightarrow f_{RC} \propto d/\varepsilon_r$ $\quad [c_{th}^{vol} = \rho c_{th}^{m}]$

Reducing $d \rightarrow C_{th} \downarrow \; C_e \uparrow$ Therefore $R_I, R_V \uparrow$ but $\Delta f \downarrow$ (3)

Best materials:

High	ρ
Low	c_{th}
Low	ε_r

R_V up to 10^3 V/W $\quad$ MEP* $\simeq 10^{-9}$ W/$\sqrt{Hz}$

<u>Applications:</u>

FTIR, Remote sensing, alarm systems

Thermal imaging

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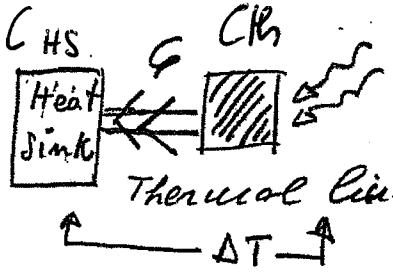
Thermal detectors

general scheme of thermal detectors →



C_{th} = thermal capacitance

G_{th} = thermal conductance

Energy balance in a time interval dt

$$P_{IN}\,dt - G_{th}\Delta T(t)\,dt = C_{th}\,d[\Delta T(t)]$$

Cauchy Poker

$$\frac{d[\Delta T(t)]}{dt} = \frac{P_{IN}}{C_{th}} - \frac{G_{th}}{C_{th}}\Delta T(t)$$

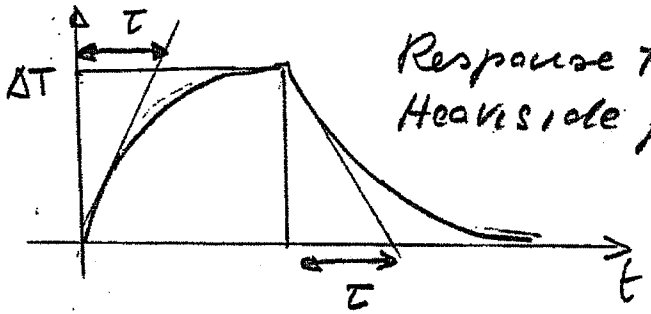
$t\to\infty$: $\frac{P_{IN}}{C_{th}} - \frac{G_{th}}{C_{th}}\Delta T = 0 \rightarrow \Delta T(\infty) = \frac{P_{in}}{G_{th}}$ output signal

Homogen. equat. $\frac{d\Delta T(t)}{dt} + \frac{G_{th}}{C_{th}}\Delta T(t) = 0$ $\quad \Delta T(t) = \Delta T_0 \exp(-t/\tau)$

$$\Delta T(t) = \frac{P_{IN}}{G_{th}}\left[1 - \exp\left(-\frac{G}{C_{th}}t\right)\right]$$

$$\Delta T(t) = \frac{P_{IN}}{G_{th}}\left[1 - \exp(-t/\tau_{th})\right]$$

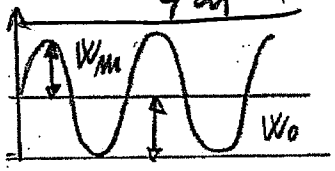
$$\boxed{\tau_{th} = \frac{C_{th}}{G_{th}}}$$



Response to the Heaviside function

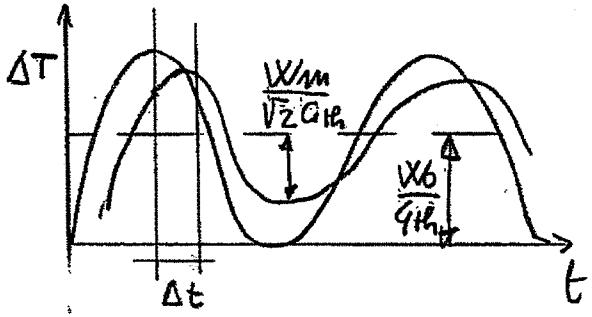
sinusoidal stimulus

$$P_{IN} = W_0 + W_m \sin(2\pi f t)$$



$t\to\infty$

$$\Delta T(t) = \frac{W_0}{G_{th}} + \frac{W_m}{G_{th}}\frac{\sin(2\pi f t + \varphi(f))}{\sqrt{1+4\pi^2 f^2 \tau_{th}^2}} \quad \left[m = \frac{W_m}{W_0}\right]$$



$f = f_c$

Cutoff frequency $\boxed{f_c = \frac{1}{2\pi\tau_{th}}}$ $\quad \omega_c = \frac{1}{\tau_{th}}$

$t\to\infty$

$$\Delta T(t) = \frac{W_0}{G_{th}} + \frac{W_m}{\sqrt{2}\,G_{th}}\sin\left(2\pi f_c t - \pi/4\right)$$

$$\Delta T^{RMS} = \frac{W_m/\sqrt{2}}{G_{th}\sqrt{1+4\pi^2 f^2 \tau_{th}^2}}$$

Increasing G_{th} improves Resp. but also τ_{th} increases

→ Better decrease Hd

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